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Efficient Stacking on Protein Amide Fragments

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The less polar π-surface of protein amide groups is exposed in many receptor binding sites, either as part of the backbone or in Gln/Asn side chains. Using quantum chemical calculations and Protein Data Bank (PDB) searches on model systems, we investigate the energetics and geometric preferences for the stacking on amide groups of a large number of heteroarenes that are relevant to medicinal chemistry. From this study, we discern that the stacking energy of an aromatic ligand substituent can be improved by: 1) orienting the fragment dipole vector such that it is aligned in an antiparallel fashion with the dipole of the interacting protein amide group, 2) increasing its dipole moment, and 3) decreasing its π-electron density. These guidelines should be helpful to more rationally exploit this interaction type in future structure-based drug design.

Introduction

Stacking interactions between π-systems are of great importance in biomolecular recognition. Experimental and theoretical studies on small, mainly aromatic model systems have provided valuable insight into the relative strength and geometrical preference of different stacking partners,[1–3] and this knowledge is routinely used in the drug design process. The prototype of π-stacking, the parallel interaction of two benzene rings, has been thoroughly investigated.[4] High-level quantum mechanical (QM) computations predict a favorable association energy of about -2.7 kcal mol^{-1} in the gas phase for the parallel-displaced configuration.[5] This geometry is also significantly enriched in the contact statistics of aromatic side chains in protein structures.[6] In contrast, the eclipsed stacking geometry is hardly observed in protein structures, and calculations reveal an approximate 1.0 kcal mol^{-1} lower energy gain than for the parallel-displaced arrangement.[5] Wang and Hobza demonstrated that the binding energy of the benzene dimer can be further increased if one of the benzene rings is replaced by an isoelectronic six-membered *N*-heteroarene.[7] Recently, significant energy differences between parallel- and antiparallel-oriented pyridine dimers were reported, highlighting the influence of dipole alignment in stacking interactions.[8]

In contrast to purely aromatic interactions, much less is known about the stacking characteristics of amide π-systems. Spectroscopic studies on γ-peptides revealed that face-to-face amide stacking with anti-aligned dipoles is competitive with intramolecular hydrogen bonding.[9,10] QM calculations on the formamide–benzene complex suggest a slightly better interaction energy than the benzene dimer.[11–13] This was rationalized by a more favorable electrostatic term due to strong dipole–quadrupole interactions which overcompensate the smaller dispersion contribution. The large experimental dipole moment of 3.7 D for *N*-methylacetamide (NMAC),[14] which we use as a model for the amide backbone fragment, suggests that stacking to this motif can be further enhanced if its partner also carries a dipole moment and is properly aligned.

Targeting amide groups with suitable ligand functionalities is attractive for two reasons: First, they are abundant in protein binding sites, either as part of the backbone or in asparagine and glutamine side chains. For a considerable proportion of these amides, the NH donor and CO acceptor groups are engaged in intra-protein hydrogen bonds,[15] while the less polar π-surface is exposed for ligand binding. Second, backbone amides are often involved in α-helix or β-sheet secondary structures, which decreases their flexibility, and thus greatly facilitates the structure-based design of a desired ligand interaction.

Examples in which stacking interactions with protein backbone amides are important for ligand binding are the S3 pocket in cathepsin L,[16] the selectivity pocket of PDE10,[17] or the S1 pocket in factor Xa.[18] Factor Xa and related chymotrypsin-like serine proteases expose in the S1 pocket two flat peptide walls, which are separated by 7–8 Å, and which preferentially bind (hetero)arenes. We previously identified two oxazole-containing factor Xa inhibitors with almost identical binding modes, but an 11-fold difference in binding affinity (Figure 1).[18] The main difference between both X-ray structures is the relative orientation of the dipoles of the oxazole linker with respect to the amide backbone of Cys191–Gln192, which is at van der Waals distance. While it is generally challenging to associate individual interactions in a protein–ligand complex with energies, it is in line with our expectation that the antiparallel dipole alignment is observed in the compound with the better binding affinity to factor Xa.

In this study, we focus on a quantitative characterization of the stacking energies and orientational preferences of various

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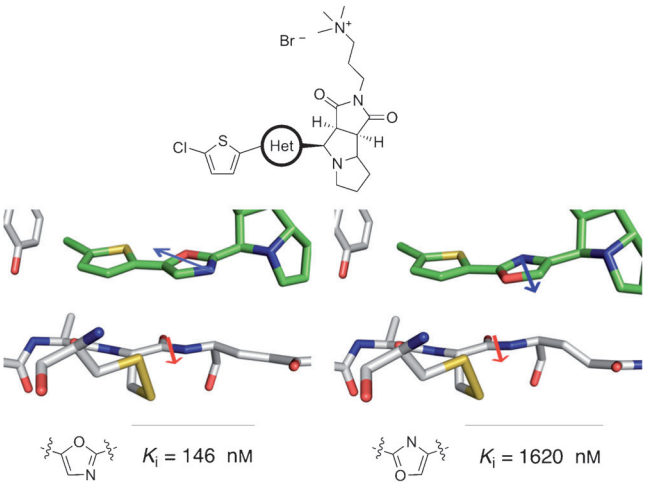

Figure 1. Inhibition constants and X-ray co-crystal structures of two oxazole-containing inhibitors with factor Xa [PDB codes: 2Y5G (left) and 2Y5H (right)].[18] Fragment dipoles of the linker heterocycle and amide backbone fragment Cys191–Gln192 are shown.

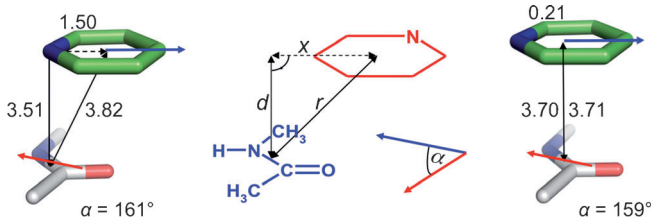

Figure 2. Coordinate system for the NMAC/heteroarene dimers, illustrated with the optimized geometries of the parallel-displaced (left) and sandwich (right) NMAC/pyridine dimers. The calculated interaction energies at the SCS-MP2/aug-cc-pVDZ level are $\Delta E = -2.69$ and -2.95 kcal mol^{-1}, respectively.

aromatic heterocycles with NMAC as a model for a planar peptide backbone fragment. We rely on quantum chemistry calculations at an adequate theoretical level and X-ray structure database mining. Examples from the Protein Data Bank (PDB), in which changes in stacking interactions with amide backbone fragments can be compared with binding affinity differences, complement this work.

Results and Discussion

The accurate computation of stacking energies of π-systems is challenging due to the large dispersion contribution, demanding a high-level theoretical treatment of electron correlation and a sufficiently large basis set.[19] Although overestimating absolute energies by 30–70%, MP2 with the aug-cc-pVDZ basis set shows an excellent correlation of relative interaction energies of stacked dimers relative to large-scale CCSD(T)/CBS calculations.[20–22] As shown by Grimme, the overestimation of absolute MP2 binding energies for weakly bonded systems can be decreased by spin-component scaling of the MP2 correlation energies (SCS-MP2).[23] This is fortunate, as the considerably lower computational cost of MP2 allows the investigation of a larger number of systems. In this study, we are primarily interested in the comparison of relative dimer interaction energies of a large number of heteroarenes with NMAC. Based on the good performance in previous studies and our own validation (Supporting Information table 1SI), which showed excellent agreement in absolute interaction energies of stacked heteroarene dimers between SCS-MP2 and CCSD(T)/CBS benchmark, we selected SCS-MP2/aug-cc-pVDZ//LMP2/cc-pVDZ as the method of choice in this study.

Influence of dipole moment vector orientation

The large dipole moment of NMAC ($\mu_{expt} = 3.7$ D) suggests that the interaction with a pyridine moiety ($\mu_{expt} = 2.2$ D) is sensitive to the relative orientation of their dipole vectors. We quantified the energy dependence in the parallel-displaced geometry (Figure 2, left) by a rotational scan of the pyridine ring in 60° increments. Figures 3 and 4 below reveal that the strongest interaction occurs close to the antiparallel configuration, while the maximum of the energy profile is located around $\alpha = 0°$. Geometry optimization without the angle constraint from the rotational scan converges to the minimum structure shown in Figure 2 left, with a binding energy of $\Delta E = -2.69$ kcal mol^{-1}. The energy difference between this minimum and the maximum of the rotational profile is 1.8 kcal mol^{-1}. We find the same overall trend for the energy profile of the NMAC/pyridine sandwich configuration (Supporting Information figure 3SI). The energy difference between the maximum and minimum in this configuration is 2.2 kcal mol^{-1}. For comparison, the energy difference between parallel- and antiparallel-oriented pyridine dimers in a parallel-eclipsed configuration is predicted to be 1.3 kcal mol^{-1} at the estimated CCSD(T) complete basis set limit.[8] An energy difference of 2.2 kcal mol^{-1} is significant; if it were a change in binding free enthalpy it would correspond to an approximate 40-fold difference in K_d values. This suggests that significant gains in binding affinity can be achieved by properly aligning stacked fragments with dipole moments in an antiparallel fashion.

According to electrostatic theory, the interaction energy of two parallel dipoles should be proportional to the magnitude of their dipole moment vectors $\vec{\mu}$, the cosine of their vector angles α, and inversely proportional to the third power of the distance from each other, r.[24] Interestingly, the shapes of the two energy curves for the parallel-displaced and sandwich NMAC/pyridine systems follow the cosine behavior derived from elementary electrostatics.

In our calculations, we find a slightly weaker interaction energy for the optimal parallel-displaced ($\Delta E = -2.7$ kcal mol^{-1}, Figure 2, left) than for the sandwich configuration ($\Delta E = -3.0$ kcal mol^{-1}, Figure 2, right). This is a qualitative difference from the interaction of more extended π-systems, such as in the benzene–benzene[5] or the benzene–triazine (Supporting Information table 1SI) dimers, in which the parallel-displaced arrangement is energetically preferred. Our absolute interaction energy for the parallel-displaced NMAC/pyridine system is only slightly weaker than for the parallel-displaced benzene–triazine dimer ($\Delta E = -3.1$ kcal mol^{-1}), calculated at the same

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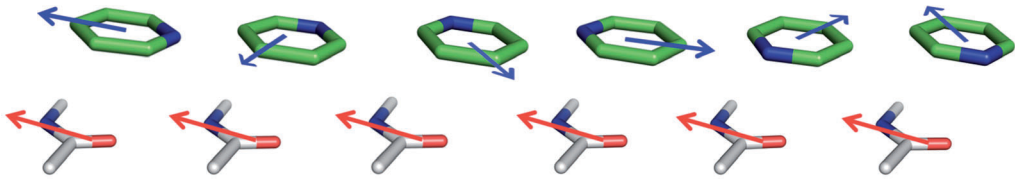


ΔE =	−0.89	−1.09	−2.03	−2.45	−2.47	−1.26 kcal mol^{-1}
d =	3.70	3.63	3.48	3.55	3.52	3.61 Å
r =	4.07	4.05	4.10	3.89	3.98	3.99 Å
x =	1.70	1.80	2.16	1.58	1.87	1.70 Å
α =	2	60	122	179	240	298°

Figure 3. SCS-MP2/aug-cc-pVDZ//LMP2/cc-pVDZ calculated dimer interaction energies and geometries for the rotational scan of the NMAC/pyridine system (parallel-displaced). For an illustration of the geometrical parameters, see the coordinate system in Figure 2. Experimental dipole moments for pyridine and NMAC are 2.2 and 3.7 D, respectively.

SCS-MP2/aug-cc-pVDZ level of theory (Supporting Information table 1SI). We speculate that the smaller dispersion contribution in the NMAC/pyridine dimer is partly compensated by a more favorable electrostatic component due to the interaction of two strong dipoles.

Early studies by Hamilton and co-workers on the stacking of thymine with an artificial receptor showed that the stacked interaction geometry could be rationalized by a high electrostatic complementarity between the partial charges of the superimposed rings.[25] We investigated whether such a simple model with local attractions/repulsions of atomic charges (δ^+/δ^-) could be used to explain the rotational profile of the NMAC/pyridine dimer. To this end, charges fitted to the MP2-calculated electrostatic potential (ESP) were determined for both monomers. In the optimized geometries of the rotational scan for the parallel-displaced association (Figure 3), the only clearly overlapping atoms are the NMAC carbonyl carbon atom and the respective pyridine ring atoms, which have ESP charges of +0.70 and −0.61 (N1), +0.40 (C2,6), −0.45 (C3,5), and +0.17 (C4). In the simple charge model, the attractive and repulsive contacts with the carbonyl C atom alternate in ~60° increments, similar to the partial charges of the pyridine ring atoms. Such behavior is not detectable in the energy profile of Figure 4, and a much better fit is achieved with the dipole interaction model.

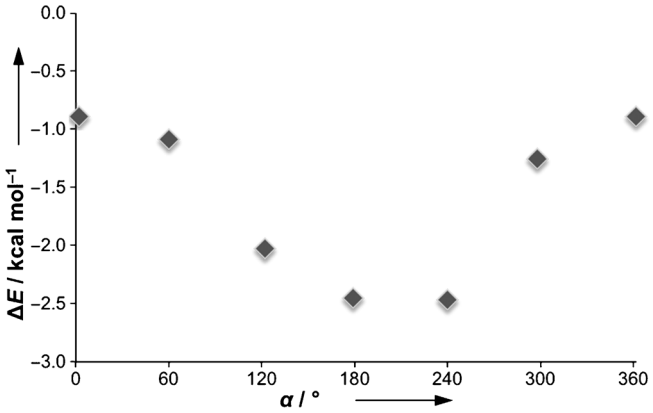
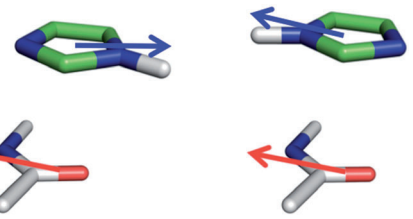


Figure 4. SCS-MP2 calculated energy profile of the rotational scan of the NMAC/pyridine system (parallel-displaced).



ΔE =	−3.58	0.52 kcal mol^{-1}
d =	3.50	3.69 Å
r =	3.71	3.81 Å
x =	1.23	0.95 Å
α =	178	4°

Figure 5. SCS-MP2/aug-cc-pVDZ//LMP2/cc-pVDZ calculated dimer interaction energies and geometries for two conformers of the NMAC/imidazole (parallel-displaced) complex with almost opposite dipole alignment. Experimental dipole moments for imidazole and NMAC are 3.8 and 3.7 D, respectively.

The influence of the dipole moment orientation on the interaction energetics is even more dramatic in the parallel-displaced NMAC/imidazole pair. Imidazole has a strong dipole moment (μ_{expt} = 3.8 D), and positioning the proton on either of the two imidazole nitrogen atoms leads to two dimer configurations in which the relative dipole moment orientation is almost inverted (Figure 5). The dimer with antiparallel dipole vectors is predicted to be strongly bound, by $\Delta E = -3.6$ kcal mol^{-1}, while the arrangement with two parallel aligned dipoles has a repulsive association energy of +0.5 kcal mol^{-1}. The interplanar distance is 0.2 Å shorter for the more favorable conformation. This correlation between interaction energy and distance is expected and generally observed in our studies.

Vertical distance scan

Apart from the rotational dependence, we probed how sensitive the binding energy reacts to changes in the vertical separation d of the parallel-displaced NMAC/pyridine dimer. Figure 6 shows the MP2 calculated energy profile in the range of 3.0–7.0 Å. The potential is quite shallow near the potential minimum (d_{min} = 3.4 Å), and substantial attraction is present even for well-separated complexes. Within the range of 3.2–3.8 Å, the binding energy is still below −2.0 kcal mol^{-1}. The equilibrium interplanar distance of this scan (3.4 Å) is slightly

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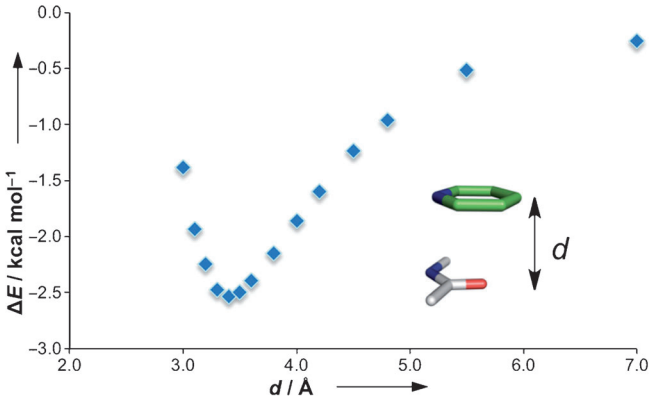

Figure 6. SCS-MP2 calculated interaction energies of the parallel-displaced NMAC/pyridine dimer as a function of the interplanar distance *d* of the monomers.

smaller than the distance observed in the optimized dimer ($d = 3.51$ Å, Figure 2, left). This is because our standard geometry optimization is at the LMP2/cc-pVDZ level, whereas the distance scan is effectively an optimization in the *d* coordinate at the SCS-MP2/aug-cc-pVDZ level.

Interaction energy of *N*-methylacetamide with different aromatic heterocycles

We selected a set of 18 different five- and six-membered heteroaromatics to explore the scope of the stacking interaction with NMAC. For each dimer, two to four starting geometries with different relative dipole moment orientations of the amide and the respective monomer were optimized, and those with the best binding energy were extracted for comparison. Relative binding energies generally show the same trend for the sandwich (Supporting Information table 3SI and figure 4SI) and parallel-displaced (Table 1, Figure 7) stacking configurations; therefore, only the latter are described in detail. In most cases, the calculated association energies for the NMAC/(hetero)arene dimers with sandwich geometry are similar to, or more negative than, the parallel-displaced orientation. Only dimers of NMAC with triazine and imidazol-2-one, respectively, showed an energetic preference for the parallel-displaced configuration. In line with the results from the rotational scan of the NMAC/pyridine dimer, the preferred arrangement of the two dipole moment vectors is antiparallel or close to it. For the parallel-displaced configurations, the optimal vector angles range from $127^\circ \leq \alpha \leq 180^\circ$ with a median of $\alpha = 161^\circ$. The smallest angles are found for the NMAC/oxazole ($\alpha = 131^\circ$) and NMAC/pyrazole ($\alpha = 127^\circ$) dimers (Supporting Information figure 5SI). The correlation between SCS-MP2/aug-cc-pVDZ and MP2/aug-cc-pVDZ calculated binding energies is excellent, with a correlation coefficient of $R^2 = 0.98$ (Supporting Information figure 6SI, left), and the correlation between the SCS-MP2/aug-cc-pVDZ and LMP2/cc-pVDZ approaches is also good ($R^2 = 0.81$, Supporting Information figure 6SI, right).

Figure 7 shows that there is a linear relationship between the strength of the dipole moment and the energy gain from the amide/aromatic stacking. The range of binding energies (−1.3 to −4.1 kcal mol^{-1}) that is covered by our set of heteroarenes is astounding. Clearly, this offers a handle for optimization of ligand affinities in protein binding sites with exposed amide groups.

It is instructive to compare the results for subsets of heterocycles in greater detail. Pyrazine, pyrimidine, and pyridazine

Table 1. Calculated binding energies and optimized geometries of various heteroarenes in complex with NMAC (parallel-displaced). From several optimized geometries of each complex, only the one with the best interaction energy is reported.[a]

Heteroarene	ΔE [kcal mol^{-1}] MP2	SCS-MP2	LMP2	r [Å]	d [Å]	x [Å]	α [°]	μ [D]
benzene	−2.50	−1.32	−0.98	3.95	3.63	1.56		0.0
pyrazine	−3.67	−2.50	−2.48	3.87	3.50	1.65		0.0
triazine	−4.91	−3.39	−4.13	3.82	3.20	2.09		0.0
furan	−2.18	−1.28	−1.23	3.83	3.59	1.33	160	0.6
thiophene	−2.55	−1.44	−1.23	3.90	3.69	1.26	162	0.5
pyrrole	−3.30	−2.12	−2.20	3.75	3.57	1.14	164	1.9
imidazole	−4.72	−3.58	−3.69	3.71	3.50	1.23	173	3.9
pyrazole	−2.82	−1.79	−1.96	3.75	3.54	1.24	127	2.3
1,2,4-triazole	−4.20	−3.11	−3.51	3.77	3.36	1.72	158	2.9
thiazole	−3.82	−2.63	−2.46	3.78	3.53	1.34	161	1.8
oxazole	−3.80	−2.74	−2.90	3.72	3.45	1.40	131	1.7
isoxazole	−3.87	−2.95	−3.11	3.79	3.41	1.65	179	3.0
1,2,5-oxadiazole	−4.89	−3.81	−4.30	3.61	3.20	1.67	159	3.3
imidazol-2-one	−4.87	−3.59	−2.09	3.71	3.40	1.47	158	4.0
pyrimidine	−4.49	−3.30	−3.94	3.85	3.40	1.80	161	2.5
pyridazine	−5.31	−4.13	−4.41	3.99	3.24	2.33	173	4.4
pyridone	−5.45	−4.01	−3.86	3.89	3.46	1.78	173	4.2
pyridine	−3.95	−2.69	−2.30	3.82	3.51	1.50	161	2.4

[a] MP2 stands for MP2/aug-cc-pVDZ//LMP2/cc-pVDZ, SCS-MP2 for the spin-component scaled MP2/aug-cc-pVDZ, and LMP2 for LMP2/cc-pVDZ. See the coordinate system in Figure 2 for an illustration of the geometrical parameters. Graphical representations of the monomers with their dipole moment vectors can be found in Supporting Information figure 1SI.

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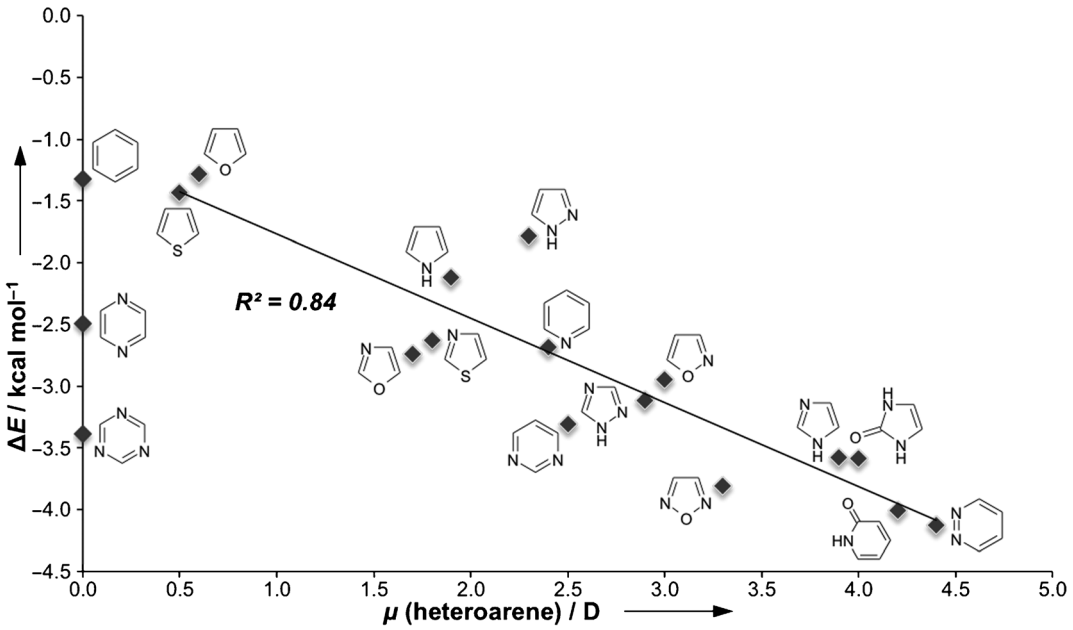

Figure 7. Plot of SCS-MP2 calculated binding energies of various NMAC/heteroarene dimers versus dipole moment of the heterocycle. The correlation coefficient is $R^2 = 0.84$.

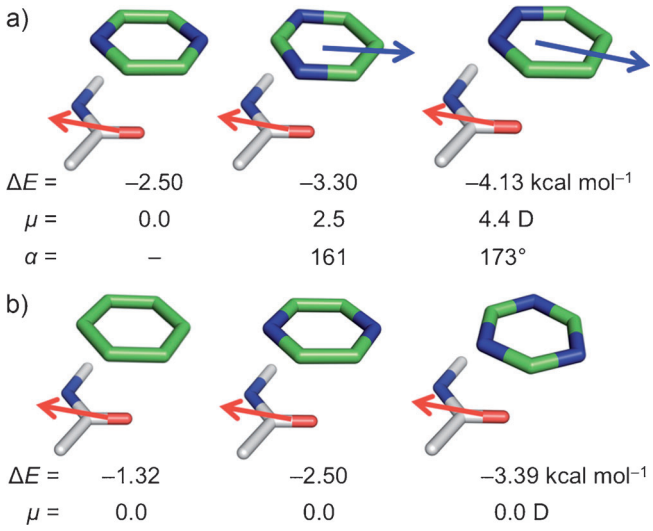

Figure 8. Comparison of SCS-MP2 calculated binding energies and dipole moment vectors of a) pyrazine, pyrimidine, and pyridazine and b) benzene, pyrazine, and triazine with NMAC.

are six-membered aromatic rings bearing two nitrogen atoms, but with drastically different dipole moments, increasing from 0.0 over 2.5 to 4.4 D (Figure 8 a). The MP2 computed interaction energy is nicely correlated with increasing dipole moment and gets more favorable by 0.8 and 1.6 kcal mol^{-1}, respectively.

Another interesting series is displayed in Figure 8 b, comparing three isoelectronic arenes (benzene, pyrazine, triazine) without a permanent dipole moment. Clearly, the interaction gets more favorable with decreasing π-electron density. From benzene to pyrazine and from pyrazine to triazine, an increase in interaction energy of 1.2 and 0.9 kcal mol^{-1}, respectively, is predicted. This increase in binding energy with increasing number of nitrogen atoms in the ring is in agreement with the results obtained by Wang and Hobza on the stacking interaction with a phenyl ring.[7]

Most of the calculations in this study were performed in the gas phase. For selected systems of Figure 7, we estimated binding energies, without counterpoise correction, in an aqueous environment using the Polarized Continuum model of Tomasi and co-workers.[26] While the results are less consistent in water than in the gas phase, the binding energies tend to be less drastically influenced by the dipole moment. This is to be expected, as fragments with higher dipole moments typically require more desolvation energy upon complexation. Effects in a less polar protein environment are more difficult to assess, but the results highlight the need to ensure that in the design process all polar ligand atoms are engaged in favorable or energetically neutral intermolecular interactions. Otherwise, the beneficial effect of stronger stacking on a protein amide group might be lost by a high desolvation penalty.

Substituent effects on dipole moment and interaction energy

Apart from switching to a different heterocycle, the orientation and strength of a fragment dipole can be modulated by hydrogen atom substitution. We used the NMAC/pyridine system to evaluate the impact of various electron-withdrawing (-F, -CF_3, -CN) and electron-donating substituents (-NMe_2) on the stacking energy. To minimize direct interactions between NMAC and the new group, the substituent was introduced *para* to the pyridine nitrogen in the most favorable parallel-displaced configuration (Figure 2, left). With these groups, it is possible to in-

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Table 2. Calculated binding energies and dipole moments of various 4-substituted pyridines in complex with NMAC (parallel-displaced).[a]

R=	H	F	CF_3	CN	NMe_2
μ [D]	2.4	0.9	0.3[b]	1.9[b]	4.4
$\Delta E_{(SCS\text{-}MP2)}$ [kcal mol^{-1}]	−2.69	−2.39	−2.74	−2.97	−5.37

[a] SCS-MP2 stands for SCS-MP2/aug-cc-pVDZ//LMP2/cc-pVDZ. [b] For R= CF_3 and CN, the dipole moment vector is inverted relative to R=H.

crease the dipole moment from 2.4 to 4.4 D (-NMe_2) or even invert the direction of the dipole moment vector (-CF_3, -CN), as shown in Table 2.

With the electron-donating dimethylamino group, the interaction energy increases drastically from −2.7 to −5.4 kcal mol^{-1}. This can be explained in part by enhanced electrostatic interactions because of the strongly increased dipole moment of the substituted pyridine ($\Delta\mu$=2.0 D). In contrast, reduction of the pyridine dipole moment or inverting its direction changes the stacking energy only slightly, and lies between −2.4 and −3.0 kcal mol^{-1}. Apparently, the mere introduction of a substituent has a stabilizing effect on dimerization. This behavior is reminiscent of the work of several research groups on substituted parallel benzene dimers, in which all substituents lead to stabilization regardless of electron-donating or electron-accepting character.[20,21,27] This was attributed to greater density overlap and greater stabilizing electron interpenetration effects in substituted arenes.[20] While a more detailed theoretical analysis for our substituted NMAC/pyridine system is out of the scope of this paper, the analogy with the work on the parallel benzene dimer is striking. In this sense, the general stabilizing effect of a substituent would be compensated by decreased dipole interactions for electron-withdrawing substituents, while for electron-donating substituents with increased dipole moment, both mechanisms lead to stabilization.

Database searches

Our calculations show a clear energetic preference for an antiparallel orientation of the dipole vectors of NMAC and pyridine. This is also observed in the PDB statistics of pyridine rings stacking on amide backbone groups (Figure 9); 63% (102 out of 161 database entries) populate vector angles α between 120 and 180° with a maximum around 165°. This experimental finding supports the results of our rotational scan calculations. Unfortunately, the PDB statistics for other heterocycles interacting with amide backbone groups are too sparse to be meaningful, and are not reported herein. Interaction statistics with the amide groups of asparagine/glutamine side chains were not attempted, as their amide orientation is often uncertain in PDB structures.[28]

While there are a number of examples in the PDB with an arene substituent of a ligand stacking on a peptide backbone, very few have published information on the SAR for arene replacement. In Supporting Information figures 7–10SI, we show four examples in which the interaction of a ligand pyridine with a protein backbone amide in an energetically favorable alignment of their dipole vectors is associated with improved binding affinity. Clearly, the observed binding affinity changes are a complex thermodynamic process and cannot be attributed to changes in the aromatic–amide interaction alone. These examples are meant as helpful illustrations and to stimulate further SAR studies in these proteins.

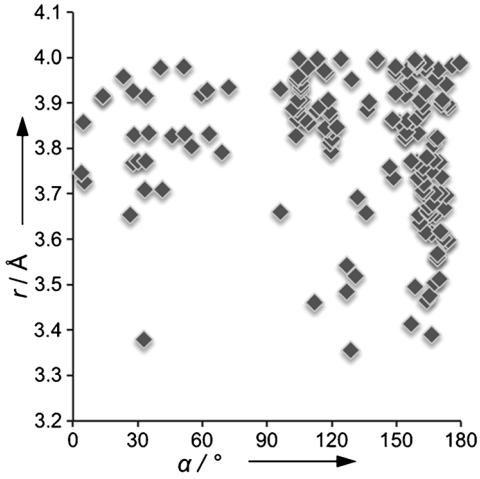

Figure 9. PDB statistics for the interaction of pyridine with dialkylated secondary amides in the peptide backbone. The *y* axis denotes the distance between the centroids of both fragments, and the *x* axis is the angle between the pyridine dipole vector and the vector between the carbonyl oxygen atom and the centroid of the amide, which is a good approximation of the true dipole moment vector (Supporting Information figure 1SI). In the statistics, only parallel examples with interplanar angles ≤30° are considered. The resolution of the PDB structures is better than 3.0 Å.

Summary and Conclusions

Amide groups are abundant in protein binding sites, and their less polar π-surface is often exposed for ligand interaction. By quantum chemical calculations on a large number of NMAC/heteroarene model systems as well as database searches, we could derive several important guidelines, which should be useful for the structure-based optimization of ligands stacking onto protein amide groups. The stacking energy of an aromatic ligand substituent can be improved by: 1) orienting the fragment dipole vector such that it is aligned in an antiparallel fashion with the dipole of the interacting protein amide group, 2) increasing its dipole moment, and 3) decreasing its π-electron density. Our calculations suggest that significant gains in binding affinity can be achieved.

The dipole moment of an aromatic ligand group can be increased by introducing or rearranging polar atoms. For ligand design, one must consider that more polar aromatic rings typically require a higher desolvation free energy to transfer them from bulk water into the receptor binding site. To avoid the beneficial stacking effect being counteracted by a significant desolvation penalty, one must ensure that all polar atoms that stack on the protein amide group are also engaged in favorable intermolecular interactions, such as hydrogen bonding, in the plane of the arene.

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In our studies, we find a good correlation between SCS-MP2 calculated dimer interaction energies and a simple dipolar interaction model. Such a model is useful, as it provides straightforward design guidelines that are easy to apply. However, it should be noted that this is an approximation to the true charge distribution and that dipole–dipole interactions are but the leading term in a multipole expansion.

Compared with other molecular interactions, there are very few examples reported in which the ligand fragment stacking on a protein amide group has been systematically varied. We are planning to address this issue by future experimental studies on selected protein binding sites as well as on well-defined model systems. We hope that our study stimulates further experimental probing and additional high-level QM calculations of this yet underexplored interaction.

Experimental Section

Computational methods

Previous studies on different arene dimers revealed that second-order Møller–Plesset perturbation theory (MP2) with an adequately large basis set and including a correction for the basis set superposition error can provide relative interaction energies very similar to benchmark coupled-cluster with singles, doubles, and non-iterative triples [(CCSD(T)] calculations.[20–22] Sanders[22] reported an excellent correlation ($R^2=0.90$) between MP2/aug-cc-pVDZ//MP2/cc-pVDZ and CCSD(T) at the estimated complete basis set (CBS) limit, and Wang and Hobza[7] showed a perfect rank correlation for different relative orientations of benzene–1,3,5-triazine complexes. In these two studies, the absolute MP2 interaction energies overestimated the binding by 30 and 70%, respectively, relative to the CCSD(T)/CBS benchmark. This deficiency of the MP2 approach can be decreased by separate scaling of parallel- and antiparallel-spin pair correlation energies (SCS-MP2).[23]

Free MP2 geometry optimization of the stacked NMAC/arene dimer converges to a T-shaped configuration,[11–13] which necessitated the use of a constraint in all our calculations to keep the two π-planes parallel. This was straightforward to define in Jaguar[29] by orienting the starting conformation of the dimers parallel to the *x*–*y* plane and constraining the distances of the four amide atoms and the non-hydrogen atoms of the heterocycle to the positive and negative values, respectively, of one optimization variable. As a consequence, we performed all geometry optimizations with Jaguar (v. 7.7) using their local MP2 (LMP2) approach.[30,31] Single-point energy calculations on the optimized geometries were done with both Jaguar (LMP2) and Gaussian 98 (MP2, SCS-MP2).[32]

Supporting Information table 1SI shows a comparison of optimized interaction energies of three previously characterized parallel benzene–1,3,5-triazine configurations[7] with our results using the MP2/aug-cc-pVDZ//LMP2/cc-pVDZ and SCS-MP2/aug-cc-pVDZ//LMP2/cc-pVDZ approaches. The MP2 method shows good agreement in absolute interaction energies with the MP2/aug-cc-pVDZ approach from Wang and Hobza,[7] and the SCS-MP2 corrected values reproduce the CCSD(T)/CBS benchmark very well. Given these results and the good performance of SCS-MP2 energy calculations in previous work, we selected SCS-MP2/aug-cc-pVDZ//LMP2/cc-pVDZ as the method of choice for our study.

Interaction energies for a set of 18 heteroarenes stacked onto NMAC were determined for relaxed parallel-displaced and sandwich configurations each. Starting geometries of the respective NMAC/heteroarene dimers were constructed from the optimized NMAC/benzene dimer using the modeling software MOLOC.[33] For each NMAC/heteroarene dimer, we took several configurations with different dipole orientations into account. LMP2 geometry optimization with the aforementioned distance constraints to keep the two π-planes parallel were run until loose convergence criteria were met, as defined in Jaguar v. 7.7. Tight convergence criteria could only be met for the energy but not the gradient changes, which remained above convergence cutoff due to the use of the parallel distance constraint. Because of this technical issue, we cannot exclude that some reported geometries might not be local minima on the potential energy surface of the heterodimers. We calculated single-point MP2 energies for the optimized heterodimers and the heteroarene monomers and eliminated the basis set superposition error in the dimer interaction energy with the counterpoise (CP) correction of Boys and Bernardi [Eq. (1)].[34]

$$\Delta E_{\text{bind}} = E_{\text{dimer}} - (E_{\text{NMAC}} + E_{\text{HetAryl}}) + \Delta E_{\text{CP}} \quad (1)$$

For the calculation of the SCS-MP2 energies, the correlation energy contributions from antiparallel- ($\alpha\beta$) and parallel-spin ($\alpha\alpha$, $\beta\beta$) electron pairs were separately scaled and added to the Hartree–Fock energy E_{HF} [Eq. (2)].

$$E_{(\text{SCS-MP2})} = E_{\text{HF}} + 6/5^{*}E_{(\alpha\beta)} + 1/3^{*}E_{(\alpha\alpha+\beta\beta)} \quad (2)$$

For the rotational scans of the NMAC/pyridine dimers, a second, torsional constraint involving N and opposite C atoms of pyridine as well as carbonyl C and O atoms of NMAC was applied. The orientation of the two dipole vectors was sampled in 60° increments from 0 to 360°. For the vertical distance scan of the parallel-displaced NMAC/pyridine dimer, an additional constraint was needed to avoid horizontal shifting at short intermolecular contacts. Here, the distance between the carbonyl carbon and the nitrogen of pyridine was set to the distance between the two planes with an additional 0.02 Å, which is the value we found for the optimized parallel-displaced dimer depicted in Figure 2, left.

The coordinate system displayed in Figure 2 is used to distinguish the dimer configurations. Here, *d* denotes the interplanar distance, and *r* is the distance between both monomer centroids. The horizontal shift can be calculated from $x=\sqrt{r^2-d^2}$. The relative orientation of the dipole moment vectors of the two monomers is given by α. For the calculation of α, the dipole vector of NMAC is approximated by the vector between the carbonyl oxygen atom and the centroid of the amide (see Supporting Information figure 1SI for the computed dipole moment vectors). While the starting conformations for the sandwich configuration had $r \cong d$, that is, $x \cong 0$ Å, the horizontal shift increased, sometimes substantially, during optimization. Only configurations with small shifts of $x \leq$ 0.7 Å were classified as sandwich in this study.

Dipole moments for all 18 investigated (hetero)arenes were calculated at the MP2/aug-cc-pVDZ level, and their strength and orientation are summarized in Supporting Information figure 1SI. They cover a large range of dipole moments from 0.0 to 4.4 D. We observed excellent agreement between computed and experimental dipole moments (Supporting Information table 2SI and figure 2SI) with a mean absolute error of 0.1 D and a maximum absolute deviation of 0.2 D. We can confirm a previous study by Bak et al., which showed that QM calculations at this level are able to accurately reproduce experimental dipole moments.[35]

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Database searches

All statistical data described in this article were derived with the ConQuest program, v. 1.14.[36] Protein–ligand interactions were analyzed by extracting binding sites in SD format from Proasis 2,[37] a curated version of the PDB, and by generating a separate database in CSD format with the program PreQuest.[38] Binding sites were extracted around all HET entries, except metals and commonly found small ions, in the PDB as of July 8, 2011. Ligands with fewer than 5 or more than 100 atoms were removed, excluding in particular large peptidic ligands. A resolution limit of 3.0 Å was applied, resulting in a database of 102 595 binding sites derived from 32 378 PDB entries.

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